

Intrinsic resolution of the MX17 “M3” reference telescope and its pointing uncertainty at the device-under-test

Four-plane cosmic self-calibration — June 2026 bench (det3 saturday long run)

MX17 cosmic-bench analysis

2026-07-15

Abstract

The MX17 “M3” reference tracker is a genuine four-plane beam telescope in each coordinate. Using a per-plane reprocessing of the cosmic data we measure the intrinsic resolution of its eight strip planes by the geometric-mean unbiased-residual method, bound the multiple-scattering (MS) contribution, and propagate the resulting *pointing uncertainty* to the device-under-test (DUT) planes at $z = 232$ and 702 mm. The reference planes resolve to $\sigma_X \approx 0.41$ mm (uniform) and $\sigma_Y \approx 0.45$ – 0.51 mm. The telescope points to the DUT with a Gaussian-core error of 0.22 mm (mid-gap, $z = 702$) to 0.27 mm ($z = 232$); this is 23–33% of the variance of the measured DUT residual. Deconvolving it gives a DUT intrinsic resolution of $\sigma_{\text{DUT}} \approx 0.40$ mm. Inside the tracking volume the reference *always* out-points the DUT in the core; a DUT can only beat the reference in extrapolation or against the MS-scattered reference tail.

1 Why the measurement is possible: geometry

The reference tracker consists of **eight MultiGen-v2 strip planes in four stations**, each station carrying one X and one Y plane (Fig. 1). Consequently each coordinate is an *independent four-point straight-line fit* — a four-plane telescope — so the classic “fit three, predict the fourth” self-calibration applies. Geometrically the stations form two tight **doublets** ($\Delta z \approx 117/120$ mm) separated by a ~ 1041 mm lever arm; the DUT slots sit *interpolated* in the gap.

2 Data and method

Data. det3 weekend “saturday” long run (resist 490 V, drift 1000 V), files 0–3, locally reprocessed after patching the tracking producer to emit the per-station cluster positions used in each fit (41 455 rays). A recomputation of $\sum \text{residual}^2$ from the emitted positions reproduces the producer’s χ^2 to a median 7×10^{-4} mm², confirming they are exactly the fit inputs. All widths below are robust **Gaussian-core** fits (iterative $\pm 2.5\sigma$) so the MS tails do not bias the intrinsic estimate; only four-hit (NClus = 4) single-track events are used.

Per-plane intrinsic resolution. For each plane k we form the *biased* residual (all four planes in the fit, width $\sigma_{\text{incl},k}$) and the *unbiased* residual (fit the other three, predict k , width $\sigma_{\text{excl},k}$). The intrinsic width is the geometric mean

$$\sigma_k = \sqrt{\sigma_{\text{incl},k} \sigma_{\text{excl},k}}, \quad \text{since} \quad \sigma_{\text{incl}}^2 = \sigma_k^2(1 - h_{kk}), \quad \sigma_{\text{excl}}^2 = \frac{\sigma_k^2}{1 - h_{kk}}, \quad (1)$$

so the leverage/geometry factor h_{kk} cancels exactly. (The alternative simultaneous unbiased-variance solve is numerically singular here, condition number $\sim 10^{16}$, because the within-doublet leave-one-out coefficients diverge — hence the geometric mean.)

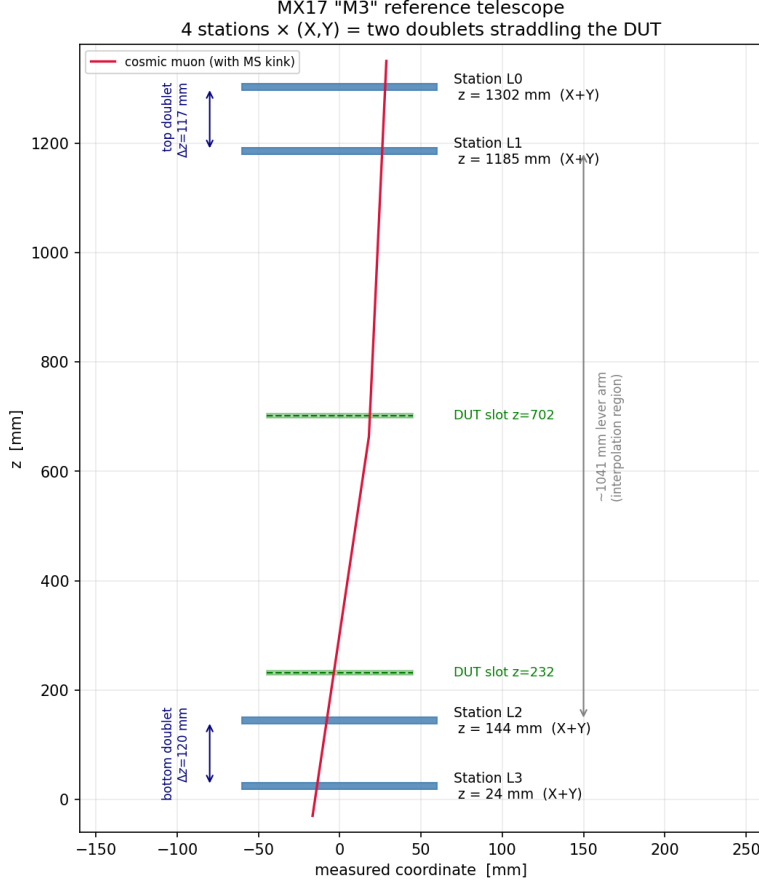


Figure 1: The M3 reference telescope: four $X+Y$ stations at $z = 24, 144, 1185, 1302$ mm forming a bottom and a top doublet, with the two DUT slots ($z = 232, 702$ mm) interpolated inside the ~ 1 mm-material gap. A cosmic muon with a small multiple-scattering kink is sketched.

Pointing. The reconstruction fits all four planes unweighted, so the predicted position at any z is $\hat{y}(z) = \sum_k g_k(z) y_k$ with purely geometric coefficients $g_k(z)$, and the pointing uncertainty is

$$P(z) = \sqrt{\sum_k g_k(z)^2 \sigma_k^2}. \quad (2)$$

The DUT intrinsic resolution follows by quadrature subtraction, $\sigma_{\text{DUT}} = \sqrt{\sigma_{\text{resid}}^2 - P(z_{\text{DUT}})^2}$.

3 Results

3.1 Per-plane intrinsic resolution

3.2 Multiple scattering

The inter-doublet kink gives $\sigma(\Delta\text{slope}) = 7.4 \text{ mrad}$ (X) against a resolution-only expectation of 6.9 mrad , i.e. an excess $\theta_{\text{MS}} \approx 2.6 \text{ mrad}$ (X ; 5.0 mrad for the noisier Y). The Highland estimate for the rough material budget ($x/X_0 \approx 0.023$) is 1.8 mrad at 1 GeV and 0.6 mrad at 3 GeV , so the measured core kink is consistent with a soft-plus-hard cosmic mix. Because the $\sim 7 \text{ mrad}$ slope resolution dominates the kink, MS is a sub-dominant term, and it enters the DUT as the

plane	σ_{incl} (μm)	σ_{excl} (μm)	σ_k (μm)	plane	σ_{incl} (μm)	σ_{excl} (μm)	σ_k (μm)
X L0	278	618	415	Y L0	343	762	511
X L1	304	553	410	Y L1	373	679	503
X L2	304	552	409	Y L2	331	600	445
X L3	275	613	411	Y L3	307	683	458

Table 1: Per-plane intrinsic resolution (geometric mean). The four X planes are uniform at $\sigma_X \approx 0.41$ mm; independent cross-check assuming a single common σ gives $411.2 \mu\text{m}$ vs the $411.3 \mu\text{m}$ geometric-mean average — agreement to $0.1 \mu\text{m}$. The Y planes are 0.45 – 0.51 mm, worse in the resistive charge-spreading direction.

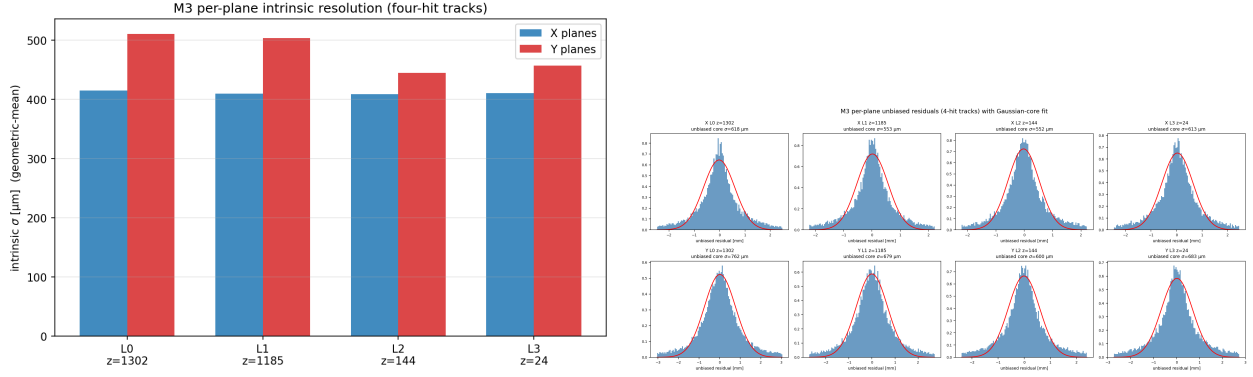


Figure 2: Left: per-plane intrinsic σ . Right: unbiased residuals with Gaussian-core fits; the peaked cores sit on non-Gaussian multiple-scattering tails.

residual *tails* (Fig. 2, right), not as core broadening. The core σ_k and $P(z)$ are therefore MS-free by construction.

3.3 Pointing at the DUT and the crossover

z (mm)	P_X (μm)	P_Y (μm)	P_{M3} (μm)	σ_{DUT} (μm)	ref. share of resid. var.
232 (bottom slot)	255	282	269	386	33 %
702 (<i>sat det3</i>)	206	242	224	413	23 %

Table 2: Reference pointing at the DUT slots (deconvolving the measured DUT core residual 0.47 mm at the $\chi^2 < 1$ & NClus = 4 recipe). Pointing is best mid-gap, near the telescope centroid $\bar{z} = 664$ mm. The deconvolved DUT intrinsic resolution is $\sigma_{\text{DUT}} \approx 0.40$ mm.

The reference pointing crosses $\sigma_{\text{DUT}} \approx 0.40$ mm only in the extrapolation regions ($z \gtrsim 1.6$ m or $z \lesssim -0.3$ m for X ; 1.37 m / -0.18 m for Y). **Inside the entire tracking volume $24 \leq z \leq 1302$ mm the reference out-points the DUT ($P \approx 0.21$ – 0.28 mm $< \sigma_{\text{DUT}}$).**

4 What this means for DUT analyses

- The M3 reference points to the DUT with a core $\sigma \approx 0.22$ mm ($z = 702$) / 0.27 mm ($z = 232$) — contributing 23/33% of the measured DUT residual *variance*. It is real but sub-dominant in the core; the DUT's own ≈ 0.40 mm dominates.

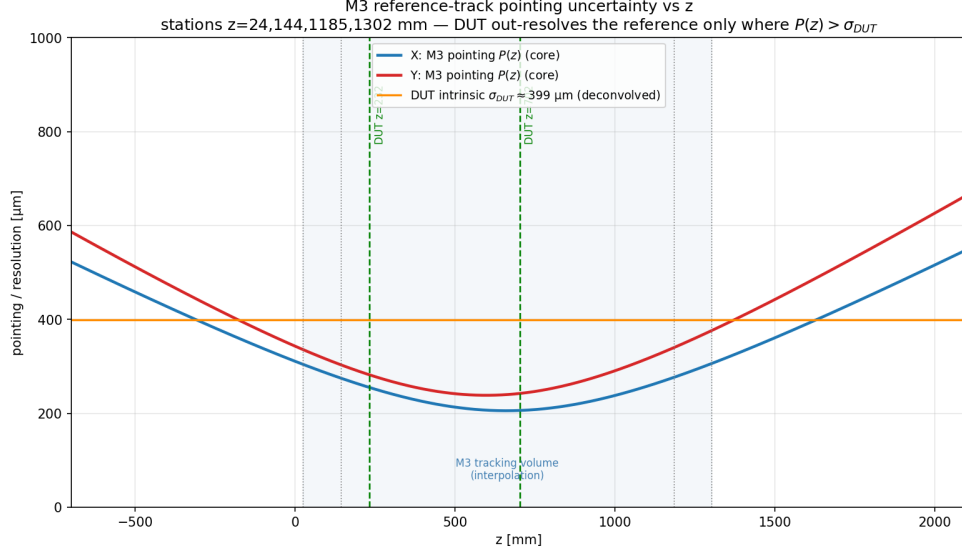


Figure 3: Reference-track pointing uncertainty $P(z)$ (Gaussian core). It is minimized near the telescope centroid and rises steeply outside the stations. The DUT out-resolves the reference only where $P(z) > \sigma_{\text{DUT}}$ (orange): above $z \approx 1.6$ m or below $z \approx -0.3$ m — i.e. never between the stations.

- Report the DUT intrinsic as $\sigma_{\text{DUT}} = \sqrt{\sigma_{\text{resid}}^2 - P_{\text{M3}}^2}$, using $P_{\text{M3}} = 0.22$ mm at $z = 702$.
- The reference is *not* the core bottleneck — its *tails* (MS-scattered, low- p muons) are. Tightening the M3 χ^2 removes those and shrinks the residual, but there is no core gain below P_{M3} ; the DUT resolution is the floor.
- In the present geometry a DUT never out-resolves the core reference between the stations. Measuring a sub-0.2 mm DUT would require a σ -weighted/GBL M3 fit, exploiting the low- P mid-gap, or an independent higher-resolution reference.

Caveats. The DUT residual (0.47 mm) is an external input; a direct join of the reprocessed rays to the det3 hits would close it self-consistently. The producer fit is unweighted, so $P(z)$ is the as-reconstructed (slightly conservative) value. θ_{MS} is a core-level estimate, not a spectrum unfold. Numbers are one run / one gas point.